



CY 247: Introduction to Molecular Thermodynamics

Course Code	CY 247
Course Name	Introduction to Molecular Thermodynamics
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B. Tech. 2nd year
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Probability, Distributions, and Equilibrium**
Distributions, Relative Probability and Fluctuations, Equilibrium, Most Probable Distribution, Le Chate-liers Principle, Equilibrium Amounts and Equilibrium Constants. [6 Lectures]
- **Energy Levels in Real Chemical Systems**
Real Chemical Reactions, The Quantized Nature of Energy, Distributions of Energy Quanta in Small Systems, Probability of a Particular Distribution of Energy, Most Probable Distribution, Energy Level Separation, Fraction of reactive Particles. [6 Lectures]
- **First Law of thermodynamics, - bonding and internal energy**
Internal Energy, Chemical Bond, Mean Bond Dissociation Energies and Internal Energy, Using Bond Dissociation Energies to Understand Chemical Reactions, The High-Energy Phosphate Bond and Other Anomalies, Beyond Covalent bond, Modern View of Bonding. [8 Lectures]
- **Entropy and the second law**
Energy Does Not Rule, Entropy Comparisons Are Informative, Standard Change in Entropy for a Chem-ical. [4 Lectures]
- **Enthalpy and the surroundings**
Enthalpy vs. Internal Energy, High Temperature Breaks Bonds. [2 Lectures]
- **Gibbs Energy and Equilibrium Constant** The Second Law - Again, Concept of equilibrium, The Low Enthalpy/High Entropy Rule, Quantitative Look at Melting Points, Vapor Pressure, Barometric Pressure, and Boiling, Isomerization Reactions, Experimental Data Can Reveal Energy Level Information, Appli-cation to Real Chemical Reactions. [8 Lectures]
- **Applications of Gibbs Energy - Phase Changes** Evaporation, Boiling, Sublimation, Vapor Deposition, Solubility, Impure Liquids. [2 Lectures]
- **Applications of Gibbs Energy – Electrochemistry** Electrical Work Is Limited by the Gibbs Energy, Gibbs Energy and Cell Potential, Actual Cell Voltages and the Nernst Equation. [6 Lectures]

Text and Reference Books

1. R. M. Hanson and S. Green, **Introduction to Molecular Thermodynamics**, University Science Books, 2008.
2. D. A. Mcquarrie and J. D. Simon, **Molecular Thermodynamics**, University Science Books, 1999.
3. Richard E. Dickerson, **Molecular Thermodynamics**, W. A. Benjamin, 1969.
4. , James W. Whalen, **Molecular Thermodynamics: A Statistical Approach**, 1991.